

Lecture 22

Eigenvalues and Eigenvectors (I)

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Outline and Review

5 mins

Roadmap of Semester

Segment 1 Solving linear systems (Lec 3-15);

Segment 2 **Lec 16-22:** Three relatively independent parts:
—Orthogonality. Least squares; orthogonal matrix (Lect 16,17,18)
—Determinant. [Important tool!] Lec 18,19, 20
—**Linear transformation.** Lec 20-21 [Advanced math perspective of matrix]

They are not directly related to solving system/problem,
but are **fundamental and useful tools**.

Segment 3 **Lec 22-27: Eigenvalues and singular values.**

Today's Lecture: Outline

Main topic: eigenvalues and eigenvectors

0. Further discussion of matrix of linear transformation
1. Definitions of eigenvalues and eigenvectors
2. How to compute: Characteristic Polynomial
3. How many eigenvalues?

Strang's book: Sec 6.1

After the lecture, you should be able to

1. Tell the definitions of eigenvalues and eigenvectors
2. Write characteristic polynomial, and compute eigenvalues and eigenvectors for small matrices
3. Tell how many eigenvalues does a matrix have

General Space: Computing Matrix of Linear Transformation

Consider a linear transformation $T: V \rightarrow W$.
Find a matrix representation of T .

Algorithm 21.1

Step 1: Pick a basis $Y = \{v_1, \dots, v_n\}$ of the domain V ;
Pick a basis $Z = \{w_1, \dots, w_m\}$ of the codomain W .

Step 2: Express $T(v_j) = a_{1j}w_1 + \dots + a_{mj}w_m \in W, \forall j$.
Write the coordinates of $T(v_j)$ as $\mathbf{a}_j = (a_{1j}, \dots, a_{mj})^T$

Step 3: Form a matrix $A = [\mathbf{a}_1, \dots, \mathbf{a}_n] \in \mathbb{R}^{m \times n}$

Conclusion: A is a matrix representation of T
with respect to the basis Y of V and the basis Z of W .

Remark: For simplicity, can also say:

A is a matrix representation of T w.r.t. (the bases) Y and Z .

Interpretation of the Matrix Representation

We get the matrix of $T(u) = u' : P_2 \rightarrow P_1$

So what?

How does this matrix help us understand the transformation?

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} = \text{matrix form of the derivative } T : u \mapsto u'.$$

Input $u = a + bx + cx^2$ $Au = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} b \\ 2c \end{bmatrix}$ Output $u' = b + 2cx$.

Coordinates of u : $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$ $\xrightarrow{A}$ $\begin{bmatrix} b \\ 2c \end{bmatrix}$ Coordinates of u'

Once we get coordinates in both spaces,
we can view the transformation T as a mapping from “**vectors to vectors**”;
and the **matrix A describes this mapping**.

Diagram: View L.T. as Operating on Coordinates

$$V \longrightarrow W$$

$$x = \sum_{i=1}^n x_i \mathbf{v}_i \mapsto \mathbf{y} = T(x) = \sum_{j=1}^m y_j \mathbf{w}_j$$

Coordinates of x

Coordinates of $y = T(x)$

$$(x_1, \dots, x_n) \xrightarrow{\text{Multiply } A} (y_1, \dots, y_n)$$

Can infer A from this relation

e.g. $\begin{bmatrix} a \\ b \\ c \end{bmatrix} \xrightarrow{\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}} (x_1, \dots, x_n)$

Once we get coordinates in both spaces,
We can view the linear transformation
as a mapping from “vectors to vectors”

Part I Matrix of Linear Transformation for Non-Euclidean Spaces (2)

15 mins

Matrix Representation of Differentiation

We know $T(\underline{u}) = u'$ defined on P_2 is a linear transformation from P_2 to P_1 .
Find a matrix representation of T .

Solution Method 2 (infer A from input-output comparison):

Step 1: Identify a basis of input space $P_2: \{1, x, x^2\}$;
a basis of output space $P_1: \{1, x\}$;

Step 2: Write the coordinates of input and output.

$$T : ax^2 + bx + c \mapsto 2ax + b.$$

$$\text{Coordinates: } \underbrace{[c, b, a]^T}_{\text{Multiply } A} \mapsto [b, 2a]^T$$

Compare: Compute $T(v_i)$
given that we know
expression of $T(u)$.

Step 3: Infer the matrix of T .

$$\begin{bmatrix} b \\ 2a \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} c \\ b \\ a \end{bmatrix}$$

Conclusion: The matrix representation of T w.r.t. Y and Z is

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Summary of This Problem Type

Type 3: (not Euclidean spaces)

Given a linear transformation T between non-Euclidean spaces .

Find a matrix representation of T .

Solution method:

T3M1: Algorithm 2 | .1

Step 1: Identify bases of input and output spaces.

Step 2: Compute the coordinates of T (each basis element of input space);

Step 3: Combine them to get A .

T3M2: [when expression of T is known and simple]

Step 1: Identify bases Y of input space and Z of output space.

Step 2: Write the expression of T , i.e.

$T(v) = w$, where v and w are expressed by bases in Step 1.

Write coordinates of $[v]_Y$ and $[w]_Z$.

Step 3: Infer a matrix A s.t. $[v]_Y = A [w]_Z$.

$$\begin{bmatrix} \vdots \\ \vdots \end{bmatrix} = \begin{bmatrix} ? & \end{bmatrix} \begin{bmatrix} \vdots \\ \vdots \end{bmatrix}$$

Example: Matrix Representation on Matrix Space

Recall: A linear transformation between $\mathbb{R}^n$ and $\mathbb{R}^m$ can be written as “matrix times sth”.

Q1: For a linear transformation between **matrix spaces**.
Can we still write it as “**matrix times sth**”?

Recall: $f: \mathbb{R}^{2 \times 2} \rightarrow \mathbb{R}^{2 \times 2}$ defined as $f(X) = X^T$ is a linear transformation.

One may think: Q1 can be rewritten as:

Q2: Can we find a matrix A s.t. $AX = X^T$ for any $X \in \mathbb{R}^{2 \times 2}$?

$\begin{matrix} \downarrow & \downarrow & \downarrow \\ 2 \times 2 & 2 \times 2 & 2 \times 2 \end{matrix}$

Example: Matrix Representation on Matrix Space

Recall: A linear transformation between $\mathbb{R}^n$ and $\mathbb{R}^m$ can be written as “matrix times sth”.

Q1: For a linear transformation between **matrix spaces**.
Can we still write it as “**matrix times sth**”?

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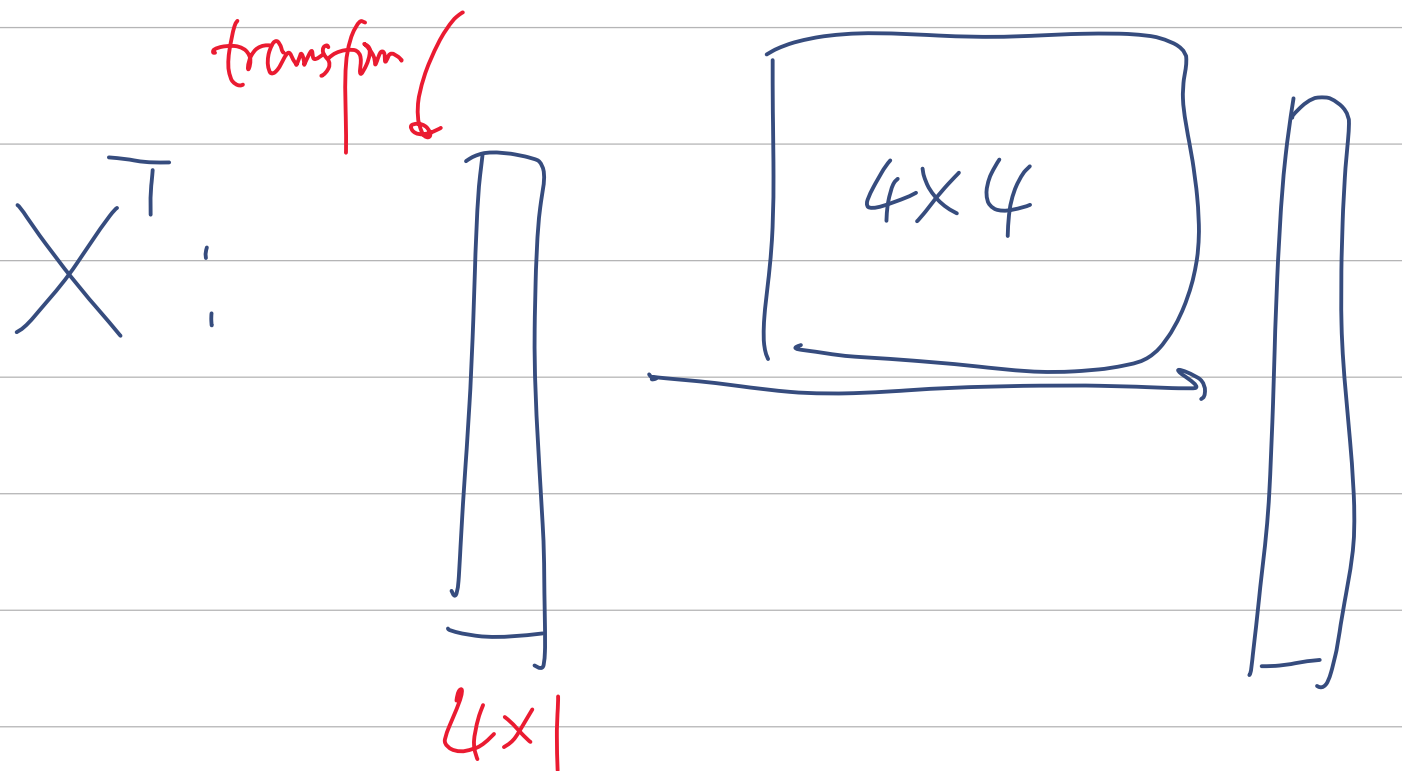
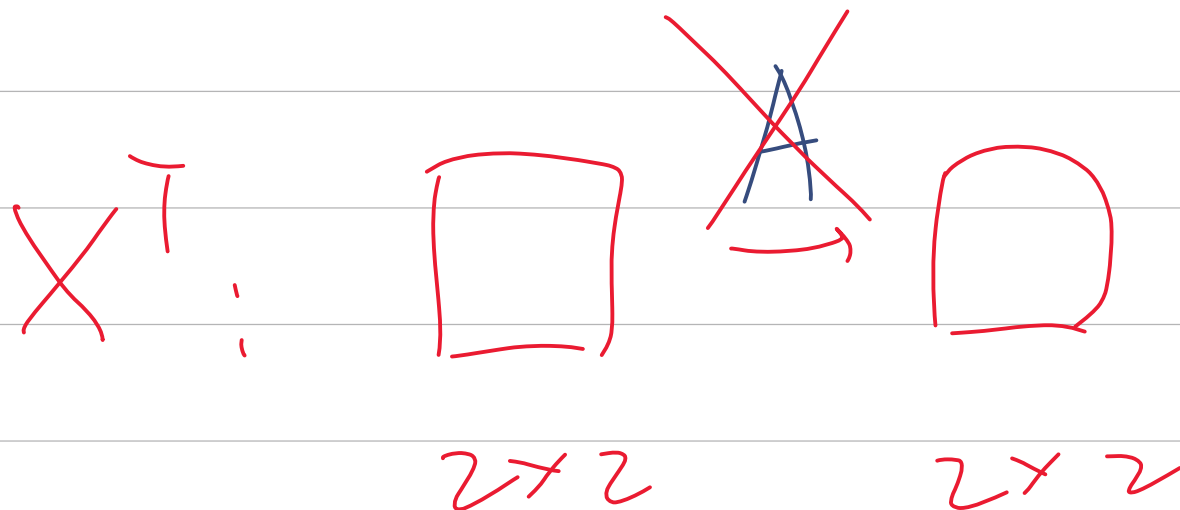
Q2: Can we find a matrix A s.t. $AX = X^T$ for any $X \in \mathbb{R}^{2 \times 2}$?

It turns out, the answer to Q2 is... **No!** [skip the derivation]

But... this is not the end.

Q2 is NOT Q1. **The answer to Q1 is “Yes”!!**

日期: /



Example: Matrix Representation of Transpose

$f: \mathbb{R}^{2 \times 2} \rightarrow \mathbb{R}^{2 \times 2}$ defined $f(X) = X^T$ is a linear transformation.
Find its matrix representation.

T3M2 (infer A from input-output comparison):

Step 1: Identify a basis of input space $\mathbb{R}^{2 \times 2}$:

The same basis for output space $\mathbb{R}^{2 \times 2}$: $Y = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

Step 2: Write the coordinates of input and output.

$$f(X) = X^T, \quad f: \begin{bmatrix} a & b \\ c & d \end{bmatrix} \mapsto \begin{bmatrix} a & c \\ b & d \end{bmatrix}$$

Coordinates: $\begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} \mapsto \begin{bmatrix} a \\ c \\ b \\ d \end{bmatrix}$

Step 3: Infer the matrix.

$$\begin{bmatrix} a \\ c \\ b \\ d \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}$$

A is a row exchange matrix in this problem

Conclusion: The matrix representation of T w.r.t. the given bases in Step 1 is

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Part II Motivation and Definitions of Eigenvalues and Eigenvectors

12 mins

“Unchanged” During Change 变化中的不变

Chemical	Physical
Before Wood log 	Before Whole lemon 
During Log getting burned 	During Lemon getting sliced 
After (new substance) Pile of ash 	After (same substance) Slices of lemon 

Chemical change:
Molecules change
[atoms are unchanged]

Physical change:
Molecules are unchanged

“Unchanged” During Change 变化中的不变

People love to understand “unchanged” parts during change.

Example in Science: physical change

Example of Chinese Ancient History (one popular theory):

“中国两千年之政，秦政也”。(谭嗣同)

Two Thousand Years of policies, just Qin Dynasty's policy.

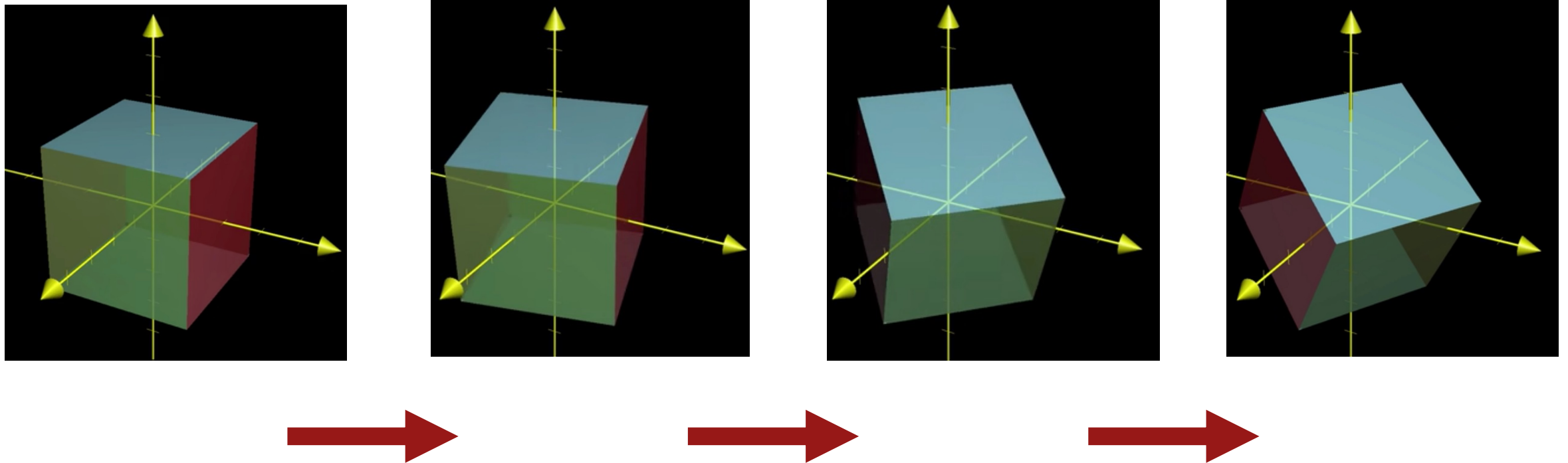
“外儒内法”。

These policies are “Confucious principles, Law Implementations”.

Why?

Better understanding the essence of things.

Example: 3D Rotation

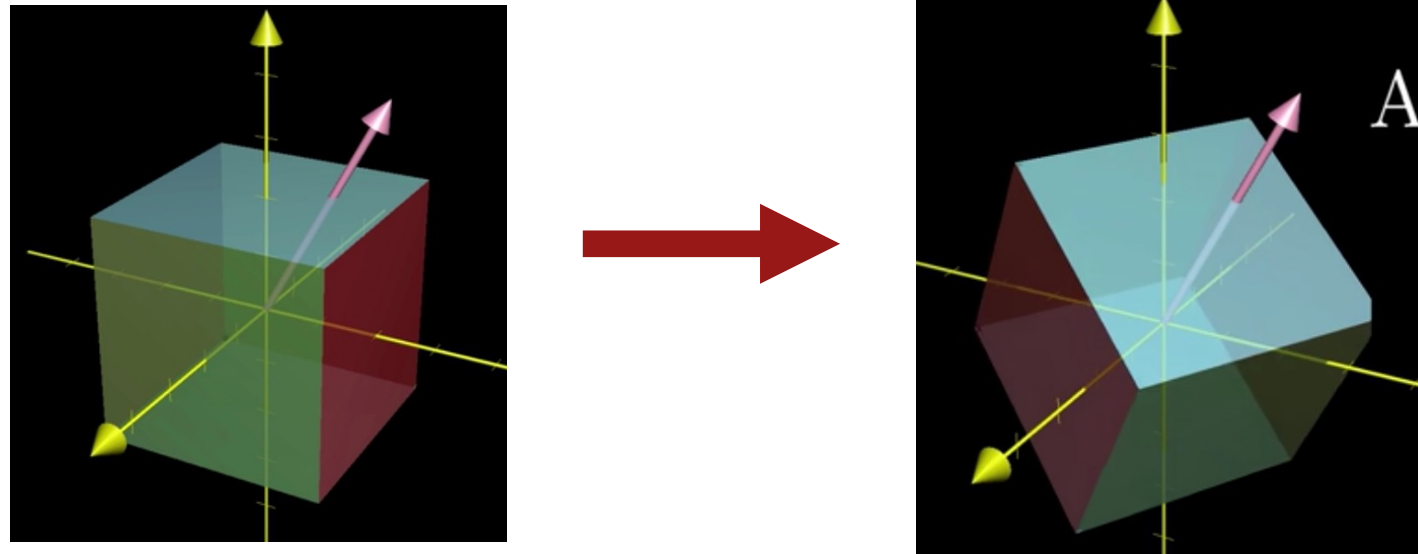


Matrix of linear transformation

$$\begin{bmatrix} \cos(\theta) \cos(\phi) & -\sin(\phi) & \cos(\theta) \sin(\phi) \\ \sin(\theta) \cos(\phi) & \cos(\theta) & \sin(\theta) \sin(\phi) \\ -\sin(\phi) & 0 & \cos(\phi) \end{bmatrix}$$

“Unchanged” During Change: Axis of Rotation

Axis of rotation: **Unchanged direction** during rotation



Rotate 30° around $\begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$
绕 $(2, 3, 1)$ 旋转 30°

V.S.

$$\begin{bmatrix} \cos(\theta) \cos(\phi) & -\sin(\phi) & \cos(\theta) \sin(\phi) \\ \sin(\theta) \cos(\phi) & \cos(\theta) & \sin(\theta) \sin(\phi) \\ -\sin(\phi) & 0 & \cos(\phi) \end{bmatrix}$$

Much simpler description

Find x and Ax on the same line?

Consider a 3×3 3D-rotation matrix A , find axis of rotation is just finding x s.t.

$$A x = x .$$

Question: Given some $A \in \mathbb{R}^{n \times n}$, can we find an $x \in \mathbb{R}^n$ such that x and Ax are on the same line?

Mathematically: $Ax = \lambda x$ for some $\lambda \in \mathbb{R}$

Eigenvalues and Eigenvectors

Definition 21.1 (Eigenvalues and Eigenvectors)

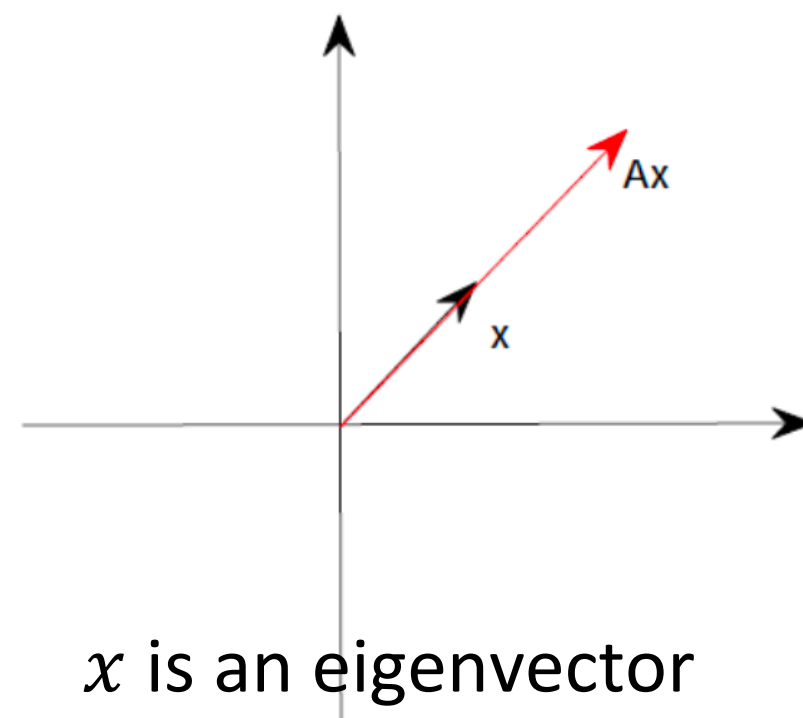
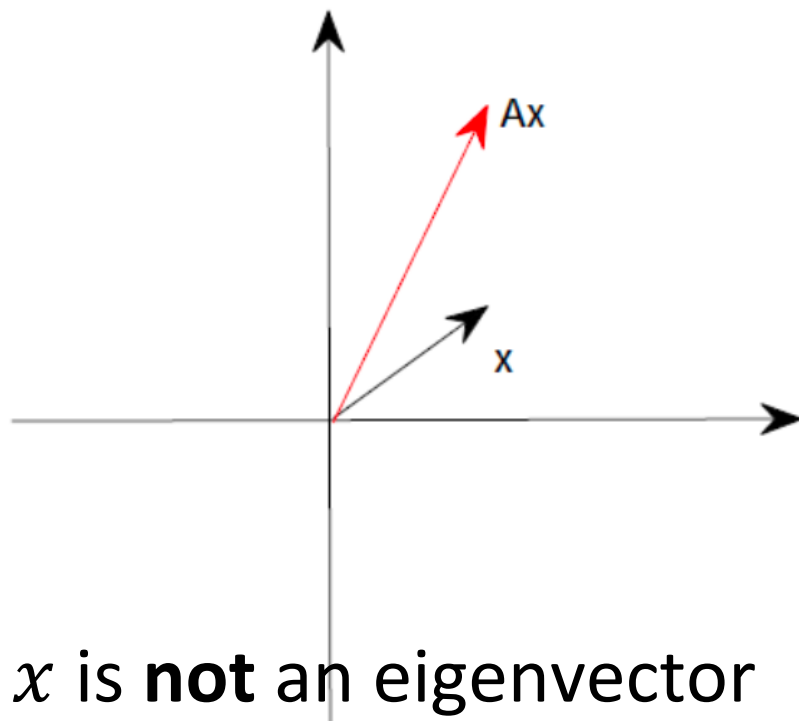
Let $A \in \mathbb{C}^{n \times n}$ be a square matrix.

If there exists a scalar λ ($\in \mathbb{R}$ or $\mathbb{C}$) and a **nonzero** vector x such that

$$Ax = \lambda x,$$

then λ is called a (real or complex) **eigenvalue** and x is called an **eigenvector** corresponding to (or associated with) λ .

Recall: $\mathbb{C} :=$ set of complex numbers, complex number $a + bi$, where $i = \sqrt{-1}$.



Example 1

Consider a 2×2 matrix $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

$$Ax = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} = x.$$

Thus $\lambda = 1$ is an *eigenvalue* of $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$.

x is an *eigenvector* of A associated with *the eigenvalue 1*.

Example 2

Consider a 2×2 matrix $A = \begin{bmatrix} 4 & -2 \\ 1 & 1 \end{bmatrix}$, $x = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$.

$$Ax = \begin{bmatrix} 4 & -2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \end{bmatrix} = 3 \begin{bmatrix} 2 \\ 1 \end{bmatrix} = 3x.$$

Thus $\lambda = 3$ is an *eigenvalue* of $A = \begin{bmatrix} 4 & -2 \\ 1 & 1 \end{bmatrix}$.

$x = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ is an *eigenvector* of A w.r.t. $\lambda = 3$.

What do Eigenvalues Represent?

Recall:

Zooming transformation, what are the eigenvalues?

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = 2 \cdot I_2$$

Geometric interpretation: The transformation $\mathbf{x} \mapsto 2\mathbf{x}$ scales all vectors uniformly by a factor of 2.

$$A\mathbf{x} = 2\mathbf{x}, \quad \forall \mathbf{x} \in \mathbb{R}^2$$

Therefore, **2 is an eigenvalue of $A = 2I_2$**

Eigenvalues: Measures the “size” of the change.

Interpretations of Eigenvalues and Eigenvectors?

Eigenvectors: “Anchor” vectors.

—Directions do not change.

(More later)

Eigenvalues: Measures the “size” of the change.

(More later)

Part III Computing Eigenvalues & Eigenvectors and Characteristic Polynomial

28 mins

Part III.1 Principles of Computation

8 mins

How to Compute Them?

We know how to **check** that a vector is an eigenvector and what is the associated eigenvalue.

How to **compute** eigenvalues and eigenvectors?

$$Ax = \lambda x$$

Question to think: Is this a linear system of equations?

How to Compute Them?

We know how to **check** that a vector is an eigenvector and what is the associated eigenvalue.

How to **compute** eigenvalues and eigenvectors?

$$Ax = \lambda x$$

Question to think: Is this a linear system of equations?

No!!! Both λ and x are unknowns.

Task: find λ and x together that satisfy $Ax = \lambda x$.

How to Simplify It?

Main Task: find $\lambda \in \mathbb{R}$ and $x \in \mathbb{R}^n \setminus \{0\}$ s.t. $Ax = \lambda x$.

$\Leftrightarrow$ find $\lambda \in \mathbb{R}$ and $x \in \mathbb{R}^n \setminus \{0\}$ s.t. $(A - \lambda I)x = 0$.

Note: For fixed λ , finding nonzero x s.t. $(A - \lambda I)x = 0$ is just solving a linear system.

Observation: $\exists$ nonzero x s.t. $(A - \lambda I)x = 0$

$\Leftrightarrow A - \lambda I$ is not invertible, [Theorem on detecting inverse]

$\Leftrightarrow \det(A - \lambda I) = 0$ [property learned recently on inverse]

Main task $\Leftrightarrow$

Subtask 1 : Find $\lambda \in \mathbb{R}$ s.t. $\det(A - \lambda I) = 0$

Subtask 2 : for such a λ , find $x \in \mathbb{R}^n \setminus \{0\}$ s.t. $(A - \lambda I)x = 0$

Part III.2 Examples

12 mins

Example 1

note: $\det(\lambda I - A) = 0 \iff \det(A - \lambda I) = 0$

Problem: Find all eigenvalues and eigenvectors of the following matrix:

$$A = \begin{bmatrix} 3 & 2 \\ 3 & -2 \end{bmatrix}$$

Step 1: Find Eigenvalues

λ should satisfy $\det(\lambda I - A) = 0$,
i.e.

$$\begin{aligned} 0 &= \det \left(\begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} - \begin{bmatrix} 3 & 2 \\ 3 & -2 \end{bmatrix} \right) \\ &= \det \begin{bmatrix} \lambda - 3 & -2 \\ -3 & \lambda + 2 \end{bmatrix} \\ &= (\lambda - 3)(\lambda + 2) - (-3)(-2) \\ &= (\lambda^2 - 3\lambda + 2\lambda - 6) - 6 \\ &= \lambda^2 - \lambda - 12 \\ &= (\lambda - 4)(\lambda + 3) \end{aligned}$$

$\Rightarrow 4, -3$ are the eigenvalues of A.

Remark: can also use:

$$ax^2 + bx + c = 0$$

the solutions are given by the **quadratic formula**:

$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Eigenvectors?

$\begin{bmatrix} 3 & 2 \\ 3 & -2 \end{bmatrix}$ has two eigenvalues 4 and -3.

After finding eigenvalues, how to find eigenvectors?

Step 2: Find eigenvectors.

Step 2.1 For the first eigenvalue.

Solve the system $(A - 4I)x = 0$

$$A - 4I = \begin{bmatrix} 3-4 & 2 \\ 3 & -2-4 \end{bmatrix} = \begin{bmatrix} -1 & 2 \\ 3 & -6 \end{bmatrix}$$

Solve $\begin{bmatrix} -1 & 2 \\ 3 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

GJ-E: $\left[\begin{array}{cc|c} -1 & 2 & 0 \\ 3 & -6 & 0 \end{array} \right] \xrightarrow{R_2+3R_1} \left[\begin{array}{cc|c} -1 & 2 & 0 \\ 0 & 0 & 0 \end{array} \right]$

Corresponding to equations: $-x_1 + 2x_2 = 0 \implies x_1 = 2x_2$

All solutions are $(x_1, x_2) = \begin{bmatrix} 2t \\ t \end{bmatrix} = t \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

All eigenvectors associated with $\lambda=4$ is $\vec{x} = t \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad t \in \mathbb{R}, t \neq 0$

$$A \cdot \vec{v} = \lambda \cdot \vec{v}$$

Eigenvectors (cont'd)

Step 2.2 For the second eigenvalue.

By definition, an eigenvector $\vec{y}$ associated with $\lambda_2 = -3$ satisfies:

$$A\vec{y} = -3\vec{y} \implies (A + 3I)\vec{y} = \vec{0}$$

$$A + 3I = \begin{bmatrix} 3+3 & 2 \\ 3 & -2+3 \end{bmatrix} = \begin{bmatrix} 6 & 2 \\ 3 & 1 \end{bmatrix}$$

$$\text{GJ-E: } \left[\begin{array}{cc|c} 6 & 2 & 0 \\ 3 & 1 & 0 \end{array} \right] \xrightarrow{R_1 \div 2} \left[\begin{array}{cc|c} 3 & 1 & 0 \\ 3 & 1 & 0 \end{array} \right] \xrightarrow{R_2 - R_1} \left[\begin{array}{cc|c} 3 & 1 & 0 \\ 0 & 0 & 0 \end{array} \right]$$

Corresponding to equations: $3y_1 + y_2 = 0 \implies y_2 = -3y_1$

All eigenvectors associated with $\lambda=4$ is

$$\vec{y} = s \begin{bmatrix} 1 \\ -3 \end{bmatrix}, \quad s \in \mathbb{R}, s \neq 0$$

Conclusion:

- The eigenvalues are $\lambda = 4$ and $\lambda = -3$.
- For $\lambda = -3$, the corresponding eigenvectors are all non-zero scalar multiples of $\begin{bmatrix} 1 \\ -3 \end{bmatrix}$.
- For $\lambda = 4$, the corresponding eigenvectors are all non-zero scalar multiples of $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$.

Algorithms to Find Eigenvalues and Eigenvectors

Algorithm 22.1

Step 1: [Compute eigenvalues]

Solve $\det(\lambda I_n - A) = 0$ to get roots $\lambda_1, \dots, \lambda_p$, where $p \leq n$.

Step 2: [Compute eigenvectors]

For each λ_i , find $N(\lambda_i I_n - A)$, the eigenspace w.r.t. λ_i ,
(any nonzero vector in $N(\lambda_i I_n - A)$ is an eigenvector corresponding to λ_i .)

The linear space spanned by eigenvectors corresponding to the same eigenvalue λ is called an **eigenspace** with respect to (w.r.t.) λ .

It is: $\{\text{all eigenvectors corresponding to } \lambda\} \cup \{0\} = \{x \in \mathbb{R}^n, Ax = \lambda x\}$.

Characteristic Polynomial

What is $\det(\lambda I_n - A)$ as a function of λ ?

$$\det(A - \lambda I) = \det \begin{pmatrix} a_{11} - \lambda & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} - \lambda & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} - \lambda \end{pmatrix} \quad \text{is a polynomial of } \lambda.$$

$\sum_i (a_{ij} - \lambda)(a_{st} - \lambda) a_{ke} \dots$ NOT $\exp(\cdot)$; NOT sine function.
each term is a polynomial

Definition 22.2 (Characteristic Polynomial)

Let $A \in \mathbb{R}^{n \times n}$ (or $\mathbb{C}^{n \times n}$) be a square matrix and λ be a variable.

Then $p_A(\lambda) = \det(\lambda I_n - A)$ is called the **characteristic polynomial** of A ,
and $p_A(\lambda) = 0$ is called the **characteristic equation**

Eigenvalues of Triangular Matrix

Claim: Eigenvalues of triangular matrices are diagonal entries.

Example:

$$A = \begin{bmatrix} 2 & 1 & 4 \\ 0 & 3 & 5 \\ 0 & 0 & -1 \end{bmatrix}$$

$$\underline{\lambda I - A} = \begin{bmatrix} \lambda - 2 & -1 & -4 \\ 0 & \lambda - 3 & -5 \\ 0 & 0 & \lambda + 1 \end{bmatrix}$$

$$\underline{\det(\lambda I - A)} = (\lambda - 2)(\lambda - 3)(\lambda + 1) \stackrel{\text{set}}{=} 0$$

$$\text{Eigenvalues: } \lambda = 2, 3, -1$$

Example 2

Example 2

Find all eigenvalues and eigenvectors of the following matrix:

$$\begin{bmatrix} 2 & -3 & 1 \\ 1 & -2 & 1 \\ 1 & 3 & -2 \end{bmatrix}$$

Example 2

Example 2

Find all eigenvalues and eigenvectors of the following matrix:

$$A = \begin{bmatrix} 2 & -3 & 1 \\ 1 & -2 & 1 \\ 1 & -3 & 2 \end{bmatrix}$$

Then characteristic equation is

$$p_A(\lambda) = \det(A - \lambda I) = \begin{vmatrix} 2 - \lambda & -3 & 1 \\ 1 & -2 - \lambda & 1 \\ 1 & -3 & 2 - \lambda \end{vmatrix} = -\lambda(\lambda - 1)^2 = 0$$

$p_A(\lambda)$ is a polynomial with degree 3.

Example 2

Example 2

$$p_A(\lambda) = \det(A - \lambda I) = \begin{vmatrix} 2 - \lambda & -3 & 1 \\ 1 & -2 - \lambda & 1 \\ 1 & -3 & 2 - \lambda \end{vmatrix} = -\lambda(\lambda - 1)^2 = 0$$

Then $\lambda_1 = 0, \lambda_2 = \lambda_3 = 1$

$$A - 0I = \begin{bmatrix} 2 & -3 & 1 \\ 1 & -2 & 1 \\ 1 & -3 & 2 \end{bmatrix} \xrightarrow{\text{Row operations}} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

The eigenspace w.r.t. 0 is $\text{Span} \left(\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right)$.

Example 2

Example 2

$$p_A(\lambda) = \det(A - \lambda I) = \begin{vmatrix} 2 - \lambda & -3 & 1 \\ 1 & -2 - \lambda & 1 \\ 1 & -3 & 2 - \lambda \end{vmatrix} = -\lambda(\lambda - 1)^2 = 0$$
$$A - I = \begin{bmatrix} 1 & -3 & 1 \\ 1 & -3 & 1 \\ 1 & -3 & 1 \end{bmatrix} \xrightarrow{\text{Row operations}} \begin{bmatrix} 1 & -3 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

The eigenspace w.r.t. 1 is $\text{Span} \left(\begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right)$.

- The eigenvalues are $\lambda = 0$ and $\lambda = 1$.

- For $\lambda = 0$, the corresponding eigenvectors are $t \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, where $t \in \mathbb{R}$ and $t \neq 0$.

- For $\lambda = 1$, the corresponding eigenvectors are $s \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$, where $s, t \in \mathbb{R}$ and $(s, t) \neq (0, 0)$.

Part III.3 Complex Eigenvalues and Eigenvectors

8 mins

Preparation: Complex Numbers

Rules for complex numbers.

For $z \in \mathbb{C} = a + ib$ where $a, b \in \mathbb{R}$, the complex conjugate is $\bar{z} = a - ib$.

- $z\bar{z} = |z|^2 = a^2 + b^2$.
- $z + \bar{z} = (a + ib) + (a - ib) = 2a \in \mathbb{R}$.
- $z - \bar{z} \in i\mathbb{R}$.
- $z = \bar{z} \implies z \in \mathbb{R}$.
- $z = -\bar{z} \implies z \in i\mathbb{R}$.
- For $z_1, z_2 \in \mathbb{C}$, $\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2$.

$$i^2 = -1.$$

Example 3: Rotation Matrix

Example 3

Find the eigenvalues and eigenvectors of the following matrix:

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

Solution:

Step 1: Compute eigenvalues.

$$\lambda I - A = \begin{bmatrix} \lambda - 0 & 0 - (-1) \\ 0 - 1 & \lambda - 0 \end{bmatrix} = \begin{bmatrix} \lambda & 1 \\ -1 & \lambda \end{bmatrix}$$

$$\det(\lambda I - A) = \lambda^2 - (-1) \cdot 1 = \lambda^2 + 1$$

So the **characteristic equation** is:

$$\lambda^2 + 1 = 0 \quad \lambda^2 = -1$$

$$\Rightarrow \lambda = i \text{ or } -i$$

$$i = \sqrt{-1}$$

Example 3: Rotation Matrix (cont'd)

Step 2: Compute eigenvectors.

For $\lambda = i$:

We solve $(A - iI)\vec{v} = 0$:

GJ-E:
$$\begin{aligned} & \text{Handwritten: } -i \cdot i = -i^2 = 1 \quad A - iI = \begin{bmatrix} -i & -1 \\ 1 & -i \end{bmatrix} \quad \text{Handwritten: } \begin{bmatrix} 1 & -i \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ & \begin{bmatrix} -i & -1 & | & 0 \\ 1 & -i & | & 0 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_1} \begin{bmatrix} 1 & -i & | & 0 \\ -i & -1 & | & 0 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_2 - R_1} \begin{bmatrix} 1 & -i & | & 0 \\ 0 & 0 & | & 0 \end{bmatrix} \end{aligned}$$

Corresponding to equations:

$$\begin{aligned} x_1 - i x_2 &= 0 \\ 0 &= 0 \end{aligned} \implies x_1 = i x_2$$

The eigenspace for $\lambda = i$ is $\text{span} \left\{ \begin{bmatrix} i \\ 1 \end{bmatrix} \right\}$. $t \cdot \begin{bmatrix} i \\ 1 \end{bmatrix}, \forall t \neq 0$.

Example 3: Rotation Matrix (cont'd)

Example 3 (restate)

Find the eigenvalues and eigenvectors of the following matrix:

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

For $\lambda = -i$:

$$A + iI = \begin{bmatrix} i & -1 \\ 1 & i \end{bmatrix}$$

Eigenspace for $-i$ is $\text{Span} \left\{ \begin{bmatrix} 1 \\ i \end{bmatrix} \right\}$

Conclusion:

- The eigenvalues are $\lambda = i$ and $\lambda = -i$.
- For $\lambda = i$, the corresponding eigenvectors are $t \begin{bmatrix} i \\ 1 \end{bmatrix}$, where $t \in \mathbb{C}$ and $t \neq 0$.
- For $\lambda = -i$, the corresponding eigenvectors are $s \begin{bmatrix} 1 \\ i \end{bmatrix}$, where $s \in \mathbb{C}$ and $s \neq 0$.

Example 4

Eigenvalues & Eigenvectors for $A = \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix}$

- Characteristic polynomial: $p_A(\lambda) = \det(A - \lambda I) = (1 - \lambda)^2 + 4$
- Characteristic equation: $(1 - \lambda)^2 + 4 = 0$, so eigenvalues are $\lambda = 1 + 2i$ and $\lambda = 1 - 2i$
- For $\lambda = 1 + 2i$, eigenvectors are $t \begin{bmatrix} -i \\ 1 \end{bmatrix}$, $t \in \mathbb{C}, t \neq 0$
- For $\lambda = 1 - 2i$, eigenvectors are $s \begin{bmatrix} i \\ 1 \end{bmatrix}$, $s \in \mathbb{C}, s \neq 0$

Part IV How Many Eigenvalues?

10 mins

How Many Eigenvalues?

Question:

How many eigenvalues does an $n \times n$ matrix have?

How many eigenvalues $\Leftrightarrow$

How many **solutions** does **$\det(\lambda I - A) = 0$** have?

Definition (Roots) 根

The roots of a single-variable function f are the solutions to the equation $f(x)=0$.

"重数"

Multiplicity of Roots

Let $f(x)$ be a polynomial.

A complex number z is a **root of f with multiplicity k** if

$$f(x) = (x - z)^k \cdot g(x), \quad \text{where } g(x) \text{ is a polynomial and } g(z) \neq 0.$$

$g(x)$ cannot be written as
 $g(x) = (x - z) h(x)$,
where h is a polynomial.

Examples:

$$f(x) = (x - 1)^3 (x + 2)^2$$

- $x = 1$ is a root of **multiplicity 3**
- $x = -2$ is a root of **multiplicity 2**

↳ roots
(count multiplicity)

$$f(x) = x^4 - 8x^3 + 26x^2 - 40x + 25 = (x - (2 + i))^2 \cdot (x - (2 - i))^2$$

- $x = 2 + i$ is a root with multiplicity **2**
- $x = \underline{2} - i$ is a root with multiplicity **2**

4 roots
(counting multiplicity)

How Many Roots Do Polynomials Have?

Linear equation: When $a \neq 0$, $ax + b = 0$ has 1 root.
(Single-variable x !)

Quadratic equation: Suppose ~~$a \neq 0$~~ .
Consider $(ax^2) + bx + c = 0$.

e.g. $x^2 - 1 = 0$ has 2 root(s)

e.g. $x^2 = 0$ has 1 root(s) or 2 roots if counting multiplicity

$\Rightarrow x=0$
 $x^2 = (x-0)^2$ e.g. $x^2 + 1 = 0$ has 0 real roots,
or 2 roots if considering complex numbers
 i & $-i$

By root formula, the equation $ax^2 + bx + c = 0$ where $a \neq 0$
has 2 complex roots, if counting multiplicity.

$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Reading: High-Order Polynomials Have ____ Roots?

Cubic equation: $ax^3 + bx^2 + cx + d = 0$ has 3 roots.

Quartic equation has 4 roots.

Root formulas were a famous problem in history.

-- 3rd and 4th order equations: found in 16th century.

≥ 5 order: a long-time mystery until 19th century.

Theorem: No solutions in radicals.

[Ruffini'1799, Abel'1824, Galois'1830]

Spectacular
Galois theory!

Root formulas: compute roots.

What about existence?

How Many Eigenvalues Does a Matrix A Have?

Theorem (Fundamental Theorem in Algebra)

Every polynomial of degree n with complex coefficients has **exactly n complex roots** (counting with multiplicity).

Remark: This is the d'Alembert–Gauss theorem in algebra

This theorem implies that $p_A(\lambda)$, as a **polynomial of degree n** , has n complex roots, i.e., **n eigenvalues** (counting with multiplicity).

Remark: The eigenvalues **may not be distinct!**

Applications of Eigenvalues and Eigenvectors

[Control] Stability in linear control theory

[Optimization and learning] Gradient descent as a Krylov subspace method

[Image processing] Image compression

[Computer science] PageRank (PCA and SVD)

[Statistics] Limit states of Markov chains

[Physics] Solutions of linear PDEs

[Physics] Cascading failure analysis

We will talk more on the data science application **Next Week!**



Summary Today (Write Your Own)

One sentence summary:

Detailed summary:

Summary Today (Instructor)

One sentence summary:

We learned eigenvalues and eigenvectors.

Detailed summary:

One **motivation**: Better characterize linear transformation.

Definition.

- $Ax = \lambda x$ for nonzero x , then λ is an **eigenvalue**, x is an **eigenvector** corresponding to λ .
- **Eigenspace**: Space spanned by all eigenvectors w.r.t. λ .

Computation.

- $\det(\lambda I_n - A)$, characteristic polynomial.
- Find eigenvalues: Solve this polynomial equation;
- Find eigenvectors: solve equation $Ax = \lambda x$ for fixed λ .

How many eigenvalues?

- Degree- n polynomial, always n complex eigenvalues (counting multiplicity)